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Unit – 2 : Session – 2 : SLO – 2 | Operational Concepts

■ Section A – Match the Following (Answers)

1. PC (Program Counter) → C. Points to the next instruction to be executed
2. MAR (Memory Address Register) → D. Holds address for accessing memory
3. MDR (Memory Data Register) → E. Holds the fetched data or instruction
4. ALU (Arithmetic Logic Unit) → A. Performs arithmetic and logic operations
5. IR (Instruction Register) → B. Temporarily holds instruction fetched from memory

■ Section B – Fill in the Blanks

- At the start of execution, the PC (PROGRAM COUNTER) is set to point to the first instruction.
- The address from the PC is loaded into the MAR, which initiates a read operation.
- The fetched instruction is placed into the IR for decoding.
- The ALU performs the operation, and the result is written back using the MAR and MDR.
- An INTERRUPT is a service request by an I/O device to the processor.

■ Section C – Activity Task: Program Execution Cycle

1. Program load via input device: the program and its data are transferred from an input device into main memory.
2. Fetch cycle (PC → MAR → Memory → MDR → IR): PC sends the instruction address to MAR; memory is read; the instruction travels through MDR into IR.
3. Decode instruction: the Control Unit examines IR, identifies the operation and the operands needed.
4. Fetch operands: operand addresses are sent to MAR and the data is read into MDR/registers.
5. ALU execution: the ALU performs the required arithmetic or logic operation on the operands.
6. Write back result: the result is sent through MDR to memory or into a destination register.
7. PC increment: PC is updated to point to the next instruction and the cycle repeats.

■ Section D – True or False

- The PC is used to perform arithmetic operations. — FALSE
- The Instruction Register (IR) holds the address of data. — FALSE (it holds the instruction itself)
- Interrupts help the processor handle I/O device requests. — TRUE
- ALU operations are performed before fetching the instruction. — FALSE
- After an interrupt is serviced, the processor resumes the program from where it was interrupted. — TRUE